

PHOTOVOLTAIC ROOF MOUNT SYSTEM

12 MODULES-ROOF MOUNTED - 5.280 kW DC, 4.608 kW AC, 925 N GRANT ST, LAFAYETTE, OR 97127

PHOTOVOLTAIC SYSTEM SPECIFICATIONS:

SYSTEM SIZE:	5.280 KW DC 4.608 KW AC
MODULE TYPE & AMOUNT:	(12) ZNSHINESOLAR ZXM7-UHLD108-440/N
MODULE DIMENSIONS:	(L/W/H) 67.79"/44.64"/1.18"
INVERTER:	(06) AP SYSTEMS DS3-L [240V]
INTERCONNECTION METHOD:	LOAD BREAKER
AHJ:	CITY OF LAFAYETTE

GOVERNING CODES:

- 2023 OREGON RESIDENTIAL SPECIALTY CODE
- 2025 OREGON STRUCTURAL SPECIALTY CODE (OSSC)
- 2023 OREGON ELECTRICAL SPECIALTY CODE (NEC 2023)
- 2025 OREGON FIRE CODE

PHOTOVOLTAIC NOTES:

1. AN INVERTER OR AN AC MODULE IN AN INTERACTIVE PHOTOVOLTAIC SYSTEM SHALL AUTOMATIC ALL DE-ENERGIZE ITS OUTPUT TO THE CONNECTED ELECTRICAL PRODUCTION AND DISTRIBUTION NETWORK UPON LOSS OF VOLTAGE IN THAT SYSTEM AND SHALL REMAIN IN THAT STATE UNTIL THE ELECTRICAL PRODUCTION AND DISTRIBUTION NETWORK VOLTAGE HAS BEEN RESTORED. (NEC 690.361)
2. ALL EXTERIOR ELECTRICAL METALLIC TUBING(EMT) CONDUIT FITTING SHALL BE RAIN TIGHT THREAD-LESS COMPRESSION TYPE.
3. MODULES AND SUPPORT STRUCTURES SHALL BE GROUNDED
4. NAMEPLATES SHALL BE PROVIDED FOR ALL CIRCUITS IN THE SERVICE DISTRIBUTION AND POWER DISTRIBUTION SWITCH BOARDS, PANEL BOARDS, DISCONNECTING SWITCHES, TERMINAL CABINETS, ETC. ALL NAMEPLATES SHALL BE PERMANENTLY ATTACHED AND BE OF SUFFICIENT CAPACITY TO WITHSTAND THE WEATHER.
5. JUNCTION BOX/COMBINER BOX HAVE TO USE COMPRESSION TYPE STRAIN RELIEF POSITIONED FOR APPROPRIATE WATER RUN OFF.
6. CONDUIT RUNS SHALL BE PROVIDED WITH SUFFICIENT WEATHERPROOF PULL BOXES OF JUNCTION BOX/COMBINER BOXES PER APPROPRIATE NEC REQUIREMENTS.
7. SEE PROVIDED CUT SHEETS FOR ADDITIONAL EQUIPMENT SPECIFICATIONS
8. WIRING MATERIALS SHALL BE SUITABLE FOR THE SUN EXPOSURE AND WET LOCATIONS. FIELD APPLIED PROTECTIVE COATINGS ARE NOT ACCEPTABLE.
9. JUNCTION, PULL AND OUTLET BOXES LOCATED BEHIND MODULES SHALL BE SO INSTALLED THAT THE WIRING CONTAINED IN THEM CAN BE RENDERED ACCESSIBLE DIRECTLY OR BY DISPLACEMENT OF MODULE(S) SECURED BY REMOVABLE FASTENERS AND CONNECTED BY A FLEXIBLE WIRING SYSTEM. (NEC 690.34)
10. IN AN UNDERGROUND PHOTOVOLTAIC SYSTEM, THE POWER SOURCE SHALL BE LABELED WITH THE FOLLOWING WARNING AT EACH JUNCTION BOX, COMBINER BOX, DISCONNECT AND DEVICE WHERE THE UNGROUNDED CIRCUITS MAY BE EXPOSED DURING SERVICE : " WARNING - ELECTRIC SHOCK HAZARD. THE CURRENT CIRCUIT CONDUCTORS OF THIS PHOTOVOLTAIC POWER SYSTEM ARE UNGROUNDED BUT MAY BE ENERGIZED WITH THE RESPECT TO GROUND DUE TO LEAKAGE PATHS AND/OR GROUND (NECE 690.35(F))
11. ALL PHOTOVOLTAIC MODULES AND ASSOCIATED EQUIPMENT AND WIRING MATERIAL SHALL BE PROTECTED FROM ANY PHYSICAL DAMAGE.
12. ALL ELECTRICAL DEVICES AND UTILIZATION EQUIPMENT SHALL BE LISTED BY AN APPROVED TESTING AGENCY.
13. OUTDOOR EQUIPMENT SHALL BE AT LEAST NEMA 3R RATED.
14. ALL SPECIFIED WIRING IS BASED ON THE USE OF COPPER
15. CONTRACTOR SHALL OBTAIN ELECTRICAL PERMITS AND SHALL COORDINATE ALL INSPECTION, COMMISSIONING AND ACCEPTANCE WITH THE CLIENT, UTILITY CO. AND CITY INSPECTORS AS NEEDED
16. DRAWINGS ARE DIAGRAMMATIC ONLY, ROUTING OF RACEWAYS SHALL BE AT THE OPTION OF THE CONTRACTOR UNLESS OTHERWISE NOTED AND SHALL BE COORDINATED WITH OTHER TRADES.
17. IF DISTANCES OF CABLE RUNS ARE DIFFERENT THAN SHOWN, THE CONTRACTOR SHALL NOTIFY ELECTRICAL ENGINEER TO VALIDATE THE WIRE SIZE. FINAL DRAWINGS WILL BE RED-LINED AND UPDATED AS APPROPRIATE.
18. WHENEVER A DISCREPANCY IN QUANTITY OF EQUIPMENT, ARISES ON THE DRAWINGS OR SPECIFICATIONS, THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING AND INSTALLING ALL MATERIAL AND SERVICES REQUIRED BY THE STRICTEST CONDITIONS NOTED ON THE DRAWINGS OR IN THE SPECIFICATIONS TO ENDURE COMPLETE COMPLIANCE AND LONGEVITY OF THE OPERABLE SYSTEM REQUIRED BY THE ARCHITECT/ENGINEER.
19. ALL BROCHURES, OPERATION MANUALS, CATALOGS, SHOP DRAWINGS, ETC. SHALL BE HANDED OVER TO THE OWNER'S REPRESENTATIVE AT THE COMPLETION OR WORK.
20. ALL WIRING CONCEALED IN WALL AND CEILING SPACES SHALL BE IN METAL CONDUIT.
21. THE SEISMIC BRACING AND ANCHORAGE OF ELECTRICAL CONDUITS SHALL BE IN ACCORDANCE

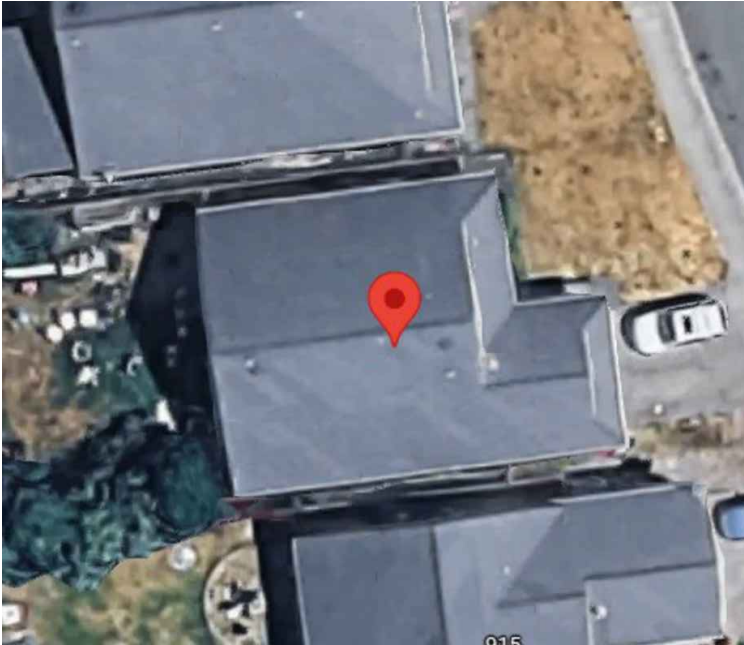
SITE DETAIL:

UTILITY:	PGE
MSP:	EXISTING 200A MSP WITH 200A MAIN BREAKER
ROOF MATERIAL:	COMPOSITE SHINGLE
ASCE 7-10 WINDSPEEDS:	(3 SEC GUST IN MPH)
WIND SPEED AND EXPOSURE:	110 MPH, C
ROOF SNOW LOAD:	20 PSF
DEAD LOAD FOR ROOF-MOUNTED PANELS ATTACHMENTS:	2.77 PSF
RACKING:	CHIKO RACKING SYSTEM FOR COMP ROOF
ARRAY AREA:	21.01 SQFT x 12 = 252.1 SQFT

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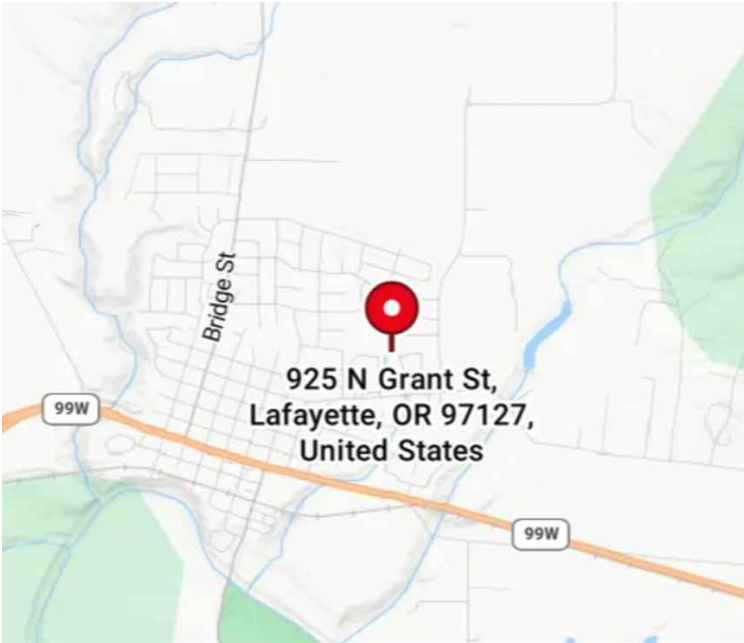


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SATELLITE VIEW

PV 0.0

SCALE: NTS



3

VICINITY MAP

PV 0.0

SCALE: NTS

1

GENERAL NOTES AND PROJECT DATA

PV 0.0



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INFINITY HOME SOLUTIONS

6405 E MILL PLAIN
VANCOUVER WA 98661
PHONE: 1-800-818-0598
INFINITY SOLAR.COM

REVISIONS

Description	Date	Rev
Initial Design	6/16/26	00

Signature with Seal

Project Name &
Address

KEVIN GONZALEZ RESIDENCE

925 N GRANT ST
LAFAYETTE, OR 97127
PHONE:- N/A
EMAIL:- N/A

Sheet Name

GENERAL NOTES
AND PROJECT
DATA

Sheet Size

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11" X 17"

Sheet Number

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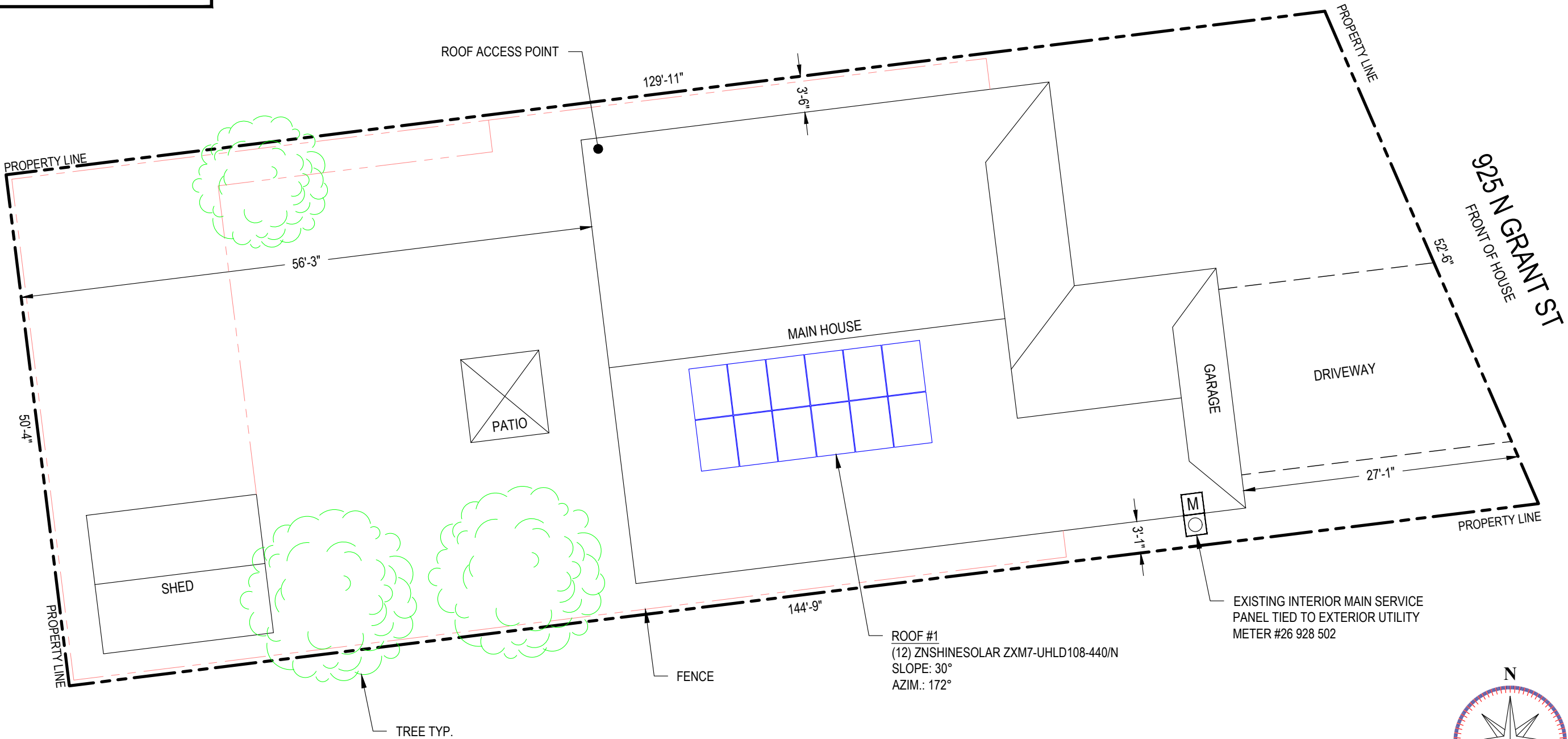
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INVERTER: (06) AP SYSTEMS DS3-L [240V]
AHJ: CITY OF LAFAYETTE

NEW PV AC DISCONNECT VISIBLE,
LOCKABLE, LABELED DISCONNECT
WITHIN 10' OF UTILITY METER

● ROOF ACCESS POINT
ROOF ACCESS POINT SHALL NOT BE LOCATED IN AREAS THAT REQUIRE THE PLACEMENT OF GROUND LADDERS OVER OPENINGS SUCH AS WINDOWS OR DOORS, AND LOCATED AT STRONG POINTS OF BUILDING CONSTRUCTION IN LOCATIONS WHERE THE ACCESS POINT DOES NOT CONFLICT WITH OVERHEAD OBSTRUCTIONS SUCH AS TREE LIMBS, WIRES OR SIGNS.





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PLOT PLAN

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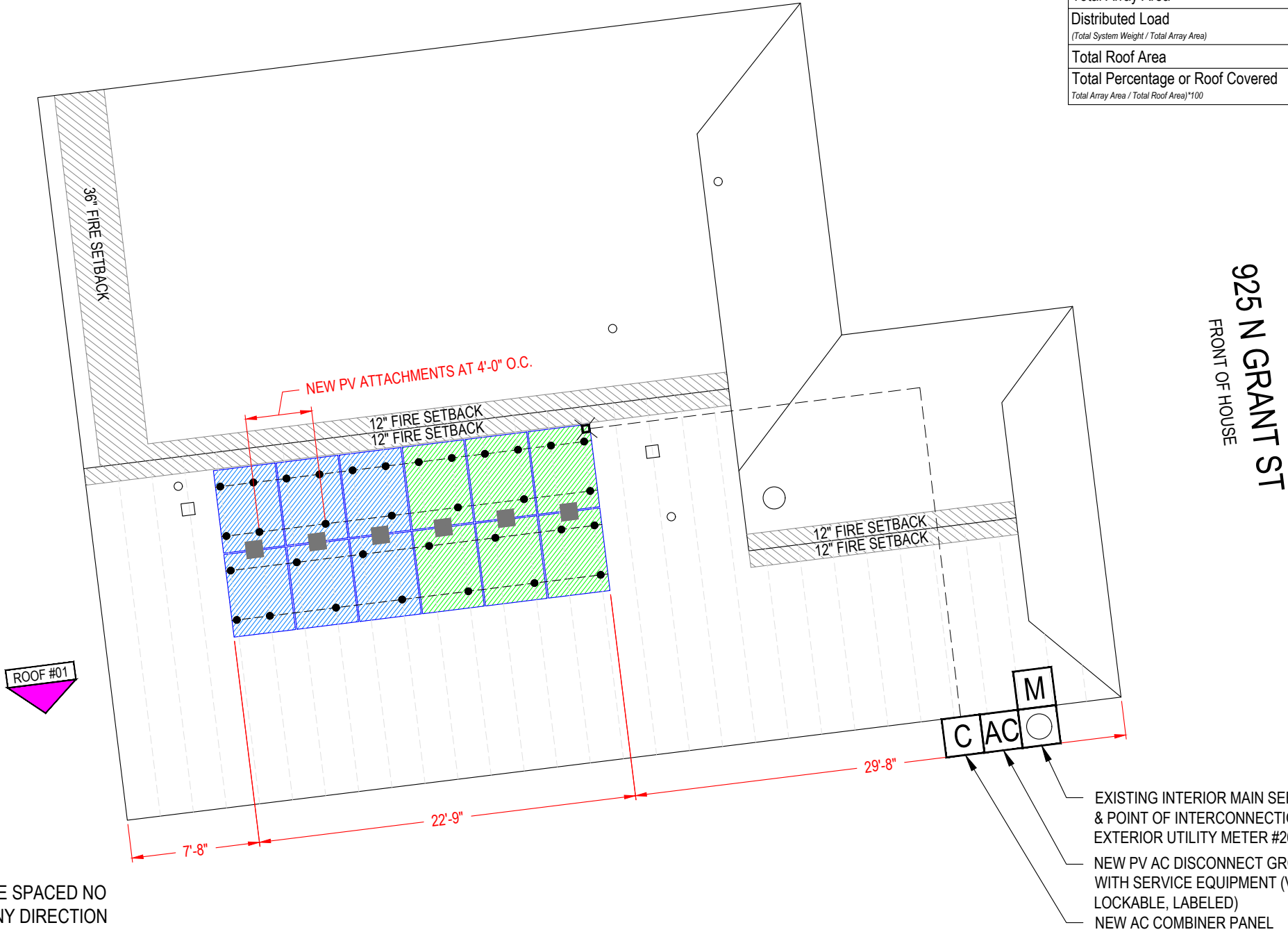
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VISIBLE, LOCKABLE, LABELED DISCONNECT
WITHIN 10' OF UTILITY METER

NOTE :
ATTIC RUN - YES
ATTIC FAN - NO
SHUTDOWN - NO

BILL OF MATERIALS		
NUMBER OF MODULES	12	ZNSHINESOLAR ZXM7-UHLD108-440/N
NUMBER OF INVERTER	06	AP SYSTEMS DS3-L [240V]
COMBINER PANEL	1	EATON BR STYLE 1-INCH LOAD CENTER 816L125RP
AC DISCONNECT	1	30A NON-FUSIBLE AC DISCONNECT, 240V
NUMBER OF ATTACHMENTS	33	CHIKO AL ROOF HOOK #167 CK-FTS-167RT2
RAILS	7	CHIKO 518 RAIL 4200MM
RAIL SPLICE	4	SPLICE KIT
MID CLAMPS	20	MID CLAMPS / UFO
END CLAMPS	8	END CLAMPS / STOPPER SLEEVE
GROUNDING LUG	4	GROUNDING LUG

MODULE, ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	12	
Module Weight	54.01	LBS
Total Module (Array) Weight	648.12	LBS
Number of Attachment point	33	
Mounting System Weight (Per Module)	1.5	LBS
Mounting System Weight	49.50	LBS
Total System Weight (Module Weight + Mounting System Weight)	697.62	LBS
Weight at Each Attachment Point (Array Weight / Number of Attachment Point)	19.64	LBS
Module Area (67.79"x44.64")	21.01	SqFt
Total Array Area	252.18	SqFt
Distributed Load (Total System Weight / Total Array Area)	2.77	Per SqFt
Total Roof Area	2344	SqFt
Total Percentage or Roof Covered Total Array Area / Total Roof Area)*100	10.76%	



NOTE :-
ROOF ATTACHMENTS SHALL BE SPACED NO GREATER THAN 24 IN. OC IN ANY DIRECTION WHERE LOCATED WITHIN 3FT. OF A ROOF EDGE, HIP, EAVE OR RIDGE.

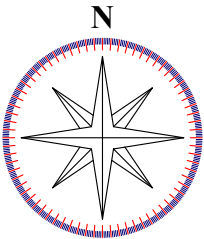
CIRCUIT(S)	
	CIRCUIT #1 - 06 MODULES WITH 03 MICROINVERTERS
	CIRCUIT #2 - 06 MODULES WITH 03 MICROINVERTERS

SYSTEM LEGEND

- EXISTING INTERIOR MAIN SERVICE PANEL & POINT OF INTERCONNECTION. TIED TO EXTERIOR UTILITY METER #26 928 502.
- NEW VISIBLE, LOCKABLE, LABELED DISCONNECT LOCATED WITHIN 10' FROM THE UTILITY METER
- NEW DEDICATED PV SYSTEM COMBINER PANEL.
- 12 NEW ZNSHINESOLAR ZXM7-UHLD108-440/N MODULES WITH NEW (06) AP SYSTEMS DS3-L [240V] INVERTERS.
- = FIRE PATHWAY
- = ROOF OBSTRUCTIONS
- = ATTACHMENT POINTS
- = RAFTER
- = RACKING SYSTEM
- = ATTIC RUN
- = CONDUIT ATTIC RUN JUNCTION BOX

ROOF SECTIONS

ROOF #01
MODULE - 12
SLOPE - 30°
AZIMUTH - 172°
MATERIAL - COMPOSITE SHINGLE
TRUSS SIZE & SPACING - 2"x4" @ 24" O.C.



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